GPUscale.net v4.9

LLM capacity & dimensioning studio: model, precision, workload and hardware in; memory fit, latency, throughput and resilient topology out.

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GPUscale.net · Untitled scenario · Kimi K3 2.8T-A50B (MoE) · weights BF16 / KV BF16 · seq 8K (+2K reasoning) · 250 concurrent, batch 4/replica · 64× worker (8 GPU) B300 288GB · TP32 · N · capacity only → 64 workers procured · 7/18/2026
```

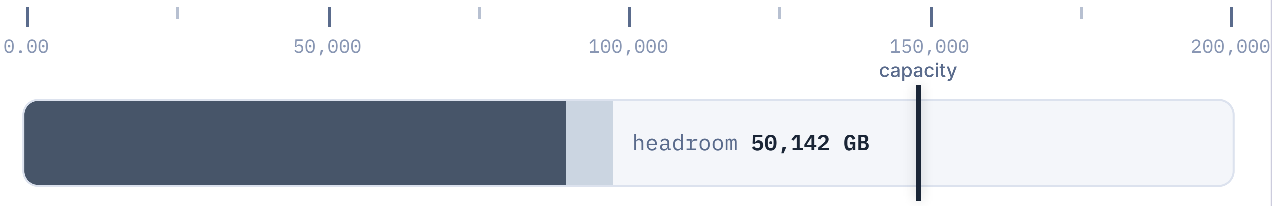
Fits & SLO targets met

97,314 GB required
147,456 GB across 512× GPU

Ready to deploy: 50,142 GB headroom, 64 of 250 calls resident in KV.

Memory ledger

scale 0–200,000 GB · 512× B300 288GB



- Weights · 16 replicas 89,600 GB 92%
- KV cache · 64 seq 15.0 GB 0%
- Activations 29.0 GB 0%
- Overhead · fixed+multi-GPU 7,670 GB 8%
- Utilization 66.0%

Time to first token

22.8 ms

prefill 8K tok on TP32 · MFU 0.5

Per-user speed

1,154 tok/s

ITL 0.87 ms/tok · batch/replica 4.00

Aggregate throughput

73,853 tok/s

64 admitted × per-user · 186 queued

Mean user latency

4.39 s

P95 ≈ 5.70 s · generates 5,000 tok

SLO compliance

targets from “Document generation”

TTFT	22.8 ms · ≤ 2,000 ms
Per-user TPS	1,154 tok/s · ≥ 30.0
P95 latency	5.70 s · ≤ 120 s

Deployment topology · workers & GPU utilization

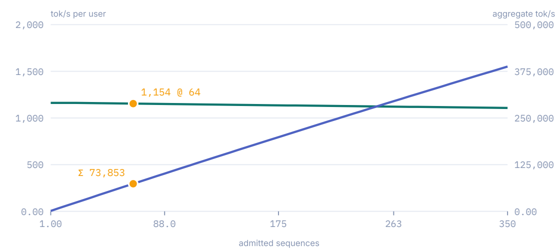
N · 16 replicas × TP32 · 66%/GPU

active worker · GPU bars show memory fill (66% of 288 GB each)

Load: **N = 64 workers · 512 GPUs** (8/worker): performance and fit are computed on these. Procured for N · capacity only: **64 workers · 512 GPUs · ≈ 614 kW** GPU TDP. No redundancy: a worker failure removes its replicas from service.

Throughput vs admitted sequences

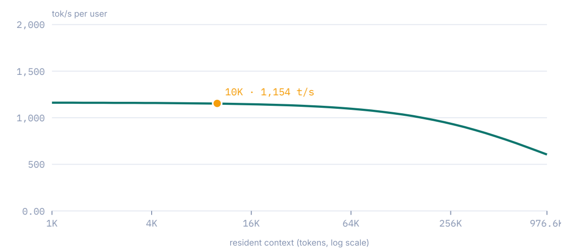
batch-aware



per-user tok/s · aggregate tok/s · memory limit · operating point

Per-user speed vs context length

at current batch



per-user tok/s · current context

Request latency anatomy

mean 4.39 s · p95 5.70 s

TTFT 22.8 ms · Overhead 30.0 ms · Reasoning 1.73 s · Visible output 2.60 s

Reading of this configuration

- ▲ TP32 spans workers (8 GPU/worker): tensor-parallel traffic leaves the NVLink domain. Either raise GPUs per worker, lower TP, or set interconnect efficiency to 0.6–0.7 to model the cross-node penalty.
- ▲ 186 of 250 calls queue at peak (only 64 admitted). Memory allows up to 13349 per replica: raise max batch toward that.
- ▸ Marginal cost of one more admitted call: 0.23 GB of KV. Speculative decoding (EAGLE-3 / MTP) would add 1.5–3× on top of the decode figures.

Recommendations

primary bottleneck: admission (batch × replicas)

Tensor parallel crosses node boundaries

TP32 with 8 GPU per worker forces TP traffic over the network. Lower TP to 8 or less, raise GPUs per worker, or model the penalty with interconnect efficiency 0.6 to 0.7.

Calls queue at peak

186 of 250 concurrent calls wait; only 64 are admitted. Memory would allow batch up to 512 per replica. Admitting everyone at batch 4 needs ≈63 workers (currently 64).

No redundancy configured

A single worker failure removes 1.56% of capacity with nothing to absorb it. N+1 costs one worker (8 GPUs, 9.60 kW) and removes that cliff.

GPUscale.net

Capacity & dimensioning studio for self-hosted LLM deployments. All figures are peak estimates; production typically achieves 70–90%. Validate with vLLM bench / GenAI-Perf before commitments.

Version `Studio 4.9.0`

Engine `v23` · library `v24 (2026-07-18)`

Scripting API at `window.GPUscale`

Private by design: everything runs in your browser. Nothing you enter is saved, uploaded or tracked. No accounts, no analytics, no cookies.